

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	SWTID-2026-7089
Project Name	Streaming IT Procurement: Automating Standard Laptop Orders With Flow Designer
Maximum Marks	4 marks

Technical Architecture:

The architecture below shows how an employee's Standard Laptop request moves through the ServiceNow cloud instance. Flow Designer runs the approval, creates and assigns the Catalog Task, updates it with the configuration details, and emails the requester. Numbers on the diagram show the order of the interactions.

Example: Automated standard laptop procurement on ServiceNow

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employees request a laptop through the Standard Laptop item in the Service Catalog; fulfillers and approvers work from their ServiceNow task and approval lists	ServiceNow Service Portal / Service Catalog (HTML, CSS, JavaScript, AngularJS widgets) on web browser or mobile
2	Application Logic-1	Approval step: once a Standard Laptop request is submitted, an approval is requested and the flow continues only after approval	ServiceNow Flow Designer (Trigger + Ask for Approval action)
3	Application Logic-2	Automatic creation of the Catalog Task, assignment to the Hardware group, and update of the task with the laptop configuration details	ServiceNow Flow Designer (Create Catalog Task, Update Record actions)
4	Application Logic-3	Sends the requester an email confirming the approved request is being processed and notifies when configuration starts	ServiceNow Flow Designer (Send Email action) and Notifications
5	Database	Stores the request, requested item and catalog task records, plus the assignment group and task details	ServiceNow platform relational database (sc_request, sc_req_item, sc_task tables)
6	Cloud Database	Database service hosted with the ServiceNow cloud instance; no separate database server is managed by the team	ServiceNow-hosted instance database
7	File Storage	No special file storage requirement; any attachments on requests or tasks are held in the platform	ServiceNow attachment storage (sys_attachment)
8	External API-1	Outbound email delivery used for requester notifications	ServiceNow email (SMTP) service

9	External API-2	Not used in the current scope; an inventory or asset system API can be added later	Not applicable
10	Machine Learning Model	Not required for this rule-based automation	Not applicable
11	Infrastructure (Server / Cloud)	Application is built and deployed directly on a ServiceNow cloud instance. Local Server Configuration: none required (browser access only). Cloud Server Configuration: ServiceNow-hosted instance	Cloud (ServiceNow instance)

Table-2 : Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Service Portal front end uses open-source UI libraries provided by the platform	AngularJS, Bootstrap
2	Security Implementations	Role-based access control (ACLs and roles), approval before fulfillment, encrypted HTTPS access, and an audit trail through request and task activity logs	ServiceNow ACLs, roles, HTTPS/TLS
3	Scalable Architecture	Multi-tier, platform-managed architecture. The flow is reusable and event-driven, so it handles additional requests without redesign; further catalog items can reuse the same logic	ServiceNow platform, Flow Designer (reusable flows/subflows)
4	Availability	Availability is provided by the ServiceNow cloud hosting; the team does not manage servers or load balancers	ServiceNow cloud infrastructure
5	Performance	The flow runs automatically on request submission, removing manual task creation and assignment, and sends notifications without delay	Flow Designer event-driven execution